

8 The curve C has equation $y = \tanh x$ for $x \geq 0$.

(a) Sketch C and state the equation of the asymptote.

[2]

(b) By considering a suitable set of N rectangles of unit width, use your sketch to show that

$$\sum_{r=1}^N \tanh r > \ln(\cosh N). \quad [3]$$





- $$S = \pi \int_{\frac{5}{4}}^{\sqrt{2}} \frac{u^2}{u^2 - 1} du. \quad [7]$$

This image shows a full page of white paper with horizontal dotted lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

(ii) Find the exact value of $\pi \int_{\frac{5}{4}}^{\sqrt{2}} \frac{u^2}{u^2 - 1} du$. You need not simplify your answer. [3]

[illegible]